Alex Doe
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EDUCATION
B.Tech in Engineering Physics, Example Institute of Technology, expected 2029
CGPA: 8.9/10

RESEARCH INTERESTS
Machine learning for high-energy physics, jet tagging, graph neural networks, anomaly detection, transformer architectures, generative models for simulation, physics-informed neural networks, scientific computing.

PROJECTS
Gated Dual-Stream Particle Transformer
Built a modified Particle Transformer for jet classification, adding a gated dual-stream attention path and multiplicative feature-space gating. Trained on a small subset of the JetClass dataset, mostly to understand the architecture.

Graph Neural Networks for Calorimeter Clustering
Implemented a GNN that groups calorimeter hits into clusters, using PyTorch Geometric. Spent most of the time on the edge-construction step.

Normalising Flows for Fast Detector Simulation
Tried a conditional normalising flow as a surrogate for a slow simulation step. Still working out why the tails are mismodelled.

SKILLS
Python, PyTorch, NumPy, ROOT, C++, Git, Linux, PyTorch Geometric, scikit-learn, pandas, Matplotlib, LaTeX, Slurm, Docker.

ACHIEVEMENTS
Department rank 3 of 120. Institute merit scholarship.